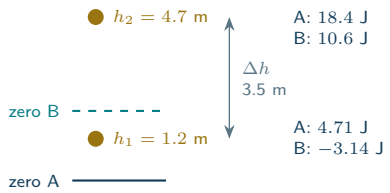
POTENTIAL ENERGY ON ONE PAGE

The potential-energy hinge

Configuration, interaction and the chosen zero, with potential kept distinct from potential energy.

1 / THE RULE AND THE PICTURE

Potential energy belongs to a configuration, and its zero is a choice



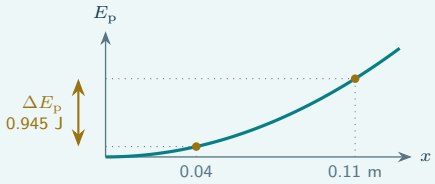
Identify the objects, the variable, the zero and the sign:

$$\boxed{\Delta E_\mathrm{p} = -W_\mathrm{cons}} \qquad \text{interaction law} + \text{geometry} + \text{reference} \to E_\mathrm{p}.$$

Lifting 0.40 kg from 1.2 m to 4.7 m gives $\Delta E_\mathrm{p} = mg\Delta h \approx 13.7$ J and $W_g = -13.7$ J. Moving the zero up by 2.0 m shifts every E_p by -7.85 J and leaves differences, work and predictions unchanged; mgh assumes a uniform local field.

2 / ELASTIC ENERGY / THE CLOSURE IS HOOKE'S LAW

$E_\mathrm{p} = \frac{1}{2}kx^2$ because the force kx grows linearly with extension

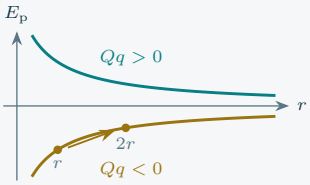


For an ideal spring with $x = 0$ the unstretched position, stretching $k = 180$ N m^{-1} from 0.04 m to 0.11 m gives $\Delta E_\mathrm{p} = \frac{1}{2}k(x_2^2 - x_1^2) = 0.945$ J, so the spring does $W_s = -0.945$ J. The work is the area under the linear force kx, hence the square, valid while Hooke's law holds. Hooke's law supplies the interaction model, energy conservation links transfers to stored energy, and $x = 0$ fixes the reference.

$$\Delta E_\mathrm{p} = \tfrac{1}{2}k(x_2^2 - x_1^2) = 0.945 \text{ J}, \qquad W_s = -0.945 \text{ J}$$

3 / RADIAL FIELDS / THE SAME $1/r$ WITH DIFFERENT SIGNS

$E_\mathrm{p} = -GMm/r$ and $E_\mathrm{p} = kQq/r$, each with zero at infinity



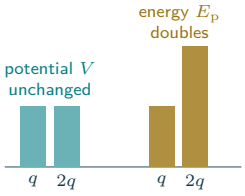
Two separated masses have zero energy and bringing them together releases energy, so the bound configuration is negative; like charges ($Qq > 0$) have positive E_p and unlike charges negative. A satellite moving from r to $2r$ rises from $-GMm/r$ to $-GMm/2r$, less negative, and two positive charges moving apart have $\Delta E_\mathrm{p} < 0$ while the electric force does positive work.

$$r \to 2r: \ \Delta E_\mathrm{p} > 0 \text{ for gravity}; \qquad \Delta E_\mathrm{p} < 0 \text{ for like charges}$$

4 / POTENTIAL AGAINST POTENTIAL ENERGY / THE MASS AND CHARGE ANALOGY

Potential belongs to the source and the place, potential energy to the pair

	gravity $(M,\ m)$	**electric** $(Q,\ q)$
field: force per unit test property	$g = F/m = GM/r^2$	$E = F/q = kQ/r^2$
potential: energy per unit test property	$V_g = E_\mathrm{p}/m = -GM/r$ in J kg^{-1}	$V_e = E_\mathrm{p}/q = kQ/r$ in J C^{-1} = V
potential energy: of the pair	$E_\mathrm{p} = mV_g = -GMm/r$	$E_\mathrm{p} = qV_e = kQq/r$
sign	$V_g < 0$, since mass attracts	V_e follows Q, E_p follows Qq



At 0.10 m from a 1.0 nC charge, $V = 90$ V for every test charge, whereas $E_\mathrm{p} = 1.8 \times 10^{-7}$ J for 2.0 nC and 3.6×10^{-7} J for 4.0 nC. At Earth's surface $V_g = -62.6$ MJ kg^{-1} for any mass, so 1 kg has $E_\mathrm{p} = -62.6$ MJ and 2 kg has -125 MJ.

$$E_\mathrm{p} = (\text{test property}) \times V\text{: } V \text{ comes from the source and the place, } E_\mathrm{p} \text{ from the pair}$$

5 / SYNTHESIS

Potential energy describes the energy of a configuration; its expression comes from interaction law

case	objects	variable	reference	closure
near-Earth gravity	Earth and object	height h	chosen level	approximately uniform field
spring	spring and object	extension x	unstretched	Hooke's law
radial gravity	two masses	separation r	zero at infinity	Newtonian gravitation
electric	two charges	separation r	zero at infinity	Coulomb's law and charge signs

ASIDE / mgh AS THE LOCAL LIMIT

$-\frac{GMm}{R+h} \approx -\frac{GMm}{R} + mgh$ with $g = GM/R^2$: mgh is the first-order term of $-GMm/r$, one interaction at two levels of approximation.

ASIDE / MISCONCEPTIONS

Potential energy belongs to the system, not an isolated object; a negative value is meaningful relative to a chosen zero; the zero can move and measurable differences cannot.

Convention. After the worksheet *The Potential-Energy Hinge*; in Python, `0.40*9.81*3.5` gives `13.73` and `0.5*180*(0.11**2-0.04**2)` gives `0.945`.

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